

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							***** MODULAR VALVES *****				
30										817	USE ONLY FOR TIE IN
30										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
30	U04		MODULAR DBB	K15AS02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
40			MODULAR DBB	K15AS06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
50	U04	VAW, 817	MODULAR DBB	K15CS06	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
60			MODULAR DBB	M15AS02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
70			MODULAR DBB	M15AS06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST
80		VAW	MODULAR DBB	M15ES02	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF SS A182 F316 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	UI2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
90		VAW, 817	MODULAR DBB	M15ZS02	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF SS A182F316 316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3		
100							***** BALL VALVES *****			VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
100		VB2								VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
110										VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
110	U04	817	BALL	A15AS05	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB LO	0	0		
120			BALL	A15CS07	4	4	BALL 1500# CF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO	0	0		
130			BALL	B15AS01	0.75	1.5	BALL 1500# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO	0	0		
140		VA5	BALL	B15AS05	2	2	BALL 1500# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB LO	0	0		
150	U04	817	BALL	B15CS07	4	4	BALL 1500# CF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO	0	0		
160							***** GLOBE VALVES *****			VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
170										VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
170			GLOBE	L15AS48	0.75	1.5	GLOBE 1500# RF A182 F316 SS TRIM 16 OS&Y HW	0	0		
180			GLOBE	L15AS41	2	2	GLOBE 1500# RF SS A351 CF8M TRIM 16 OS&Y HW	0	0		
190	U04	817	GLOBE	L15CS42	4	4	GLOBE 1500# CF SS A351 CF8M TRIM 16 OS&Y GO	0	0		
200							***** CHECK VALVES *****				
200											
210											
210		VC1	CHECK	C15AS50	0.75	1.5	CHECK PISTON 1500# RF SS A182 F316 TRIM 16	0	0		
220	U04		CHECK	C15AS45	2	2	CHECK DUAL WAFER LUG 1500# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0		
230	U04	817	CHECK	C15CS52	4	4	CHECK DUAL DBL FLG 1500# CF SS A351 CF8M TRIM 16	0	0		
240							***** PIPE *****				
240											
250											
250			PIPE	5365207	0.75	1.5	PIPE SCH 160 SMLS SS A312 TP316/316L PE	0	0		
260			PIPE	5365205	2	6	PIPE SCH 160 SMLS SS A312 TP316/316L BE	0	0		
270			PIPE	5365206	0.75	4	PIPE SCH XXS SMLS SS A312 316/316L SS BE	0	0		
280							***** FITTINGS *****				
280											
290											
290	U04		ELBOW	5449140	6	6	ELL BW 90 DEG LR SCH 160 SS A403 WP316/316L-S	0	0		
300			ELBOW	5449140	0.75	2	ELL BW 90 DEG LR SCH 160 SS A403 WP316/316L-S	0	0		
310			ELBOW	5449142	0.75	2	ELL BW 45 DEG LR SCH 160 SS A403 WP316/316L-S	0	0		
320											
320			TEE	5449190	0.75	2	TEE SCH 160 SS A403 WP316/316L-S	0	0		
330			FLANGE	5449686	0.75	1.5	TEE RED SCH 160 316/316L SS A403	1	2		
340											
340			CAP	5449055	0.75	2	CAP SCH 160 SS A403 WP316/316L-S	0	0		
350											
350	U04		RED CONC	5448912	1.5	2	REDUCER CONC SCH XXS/160 SS A403 WP 316/316L-S	3	4		
360			RED CONC	5449689	0.75	1	REDUCER CONC SCH 160 SS A403 WP 316/316L-S	1	2		
370	U04		RED ECC	5449187	1.5	2	REDUCER ECC SCH XXS/160 SS A403 WP316/316L-S	3	4		
380			RED ECC	5449690	0.75	1	REDUCER ECC SCH 160 SS A 403 WP 316/316L-S	1	2		
390	U04		RED ECC	5449180	3	3	REDUCER ECC SCH 160/SCH XXS SS A403 WP316/316L-S	6	6		
400											
400			OLETS	5504047	0.75	1.5	NIPOFLANGE 1500# RF SCH 160 SS A182 F316/316L	2	2		
410		UI2	OLETS	5504048	0.75	2	NIPOFLANGE 1500# RF SCH XXS/160 SS A182 F316/316L	4	4		
420											
420	U04	UI2	ELBOFLG	5516439	1.5	1.5	ELBOFLANGE 1500# RF SCH XXS/160 A182 F316/316L	4	4		
430							*****OLETS*****				
430											
440											
440			OLETS	5537231	0.75	1.5	SOCKOLET 6000# SS A182 F316/316L	2	2		
450			OLETS	5533236	0.75	1.5	SOCKOLET FLAT BOTTOM 6000# SS A182 F316/316L	0	0		
460											
460		UD7	OLETS	5515772	0.75	0.75	THREDOLET 6000# SS A182 F316/316L	2	2		
470											
470	U04	817	OLETS	5509668	0.75	0.75	WELDOLET SCH 160 SS A182 F316/316L	2	2		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES BUTT - WELDED SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 1 MM	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -50 / 200°C	12/16/2024	BY OZAC	U04	Page 1 of 2	1500P1

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480							***** COUPLINGS, UNIONS & PLUGS *****			
480										
490										
490			COUPLING	5480172	0.75	1.5	COUPLING 6000# SW SS A182 F316/316L	0	0	
500			FLANGE	5480175	0.75	1	COUPLING RED 6000# SW SS A182 F316/316L	1	1.5	
510										
510			UNION	5480176	0.75	1.5	UNION 6000# SW SS A182 F316/316L	0	0	
520										
520		UD7	FITTING SCRD	5498383	0.75	0.75	PLUG ROUND HEAD SCRD SS A182 F316/316L	0	0	
530										
530		816					***** FLANGES *****			
540										
540	U04	FA8	FLANGE	5600664	0.5	0.75	FLG SCRD (FNPT) REDUCING 1500# RF SS A182 F316/316L	0.75	1.5	
550										
550			FLANGE	5851205	0.75	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
560										
560			FLANGE	5583003	0.75	6	FLG WN 1500# RF SS A182 F316/316L SCH 160 BORE	0	0	
570	U04		FLANGE	5583004	0.75	2	FLG WN 1500# RTJ SS A182 F316/316L SCH 160 BORE	0	0	
580	U04	817	FLANGE	5619840	4	4	FLG COMPACT 1500# RJ SS A182 F316/316L SCH XXS LINE CLASS 1500P1	0	0	
590	U04		FLANGE	5584177	0.75	2	FLG WN 2500# RTJ SS A182 F316/316L SCH 160	0	0	
600			BLIND	5616564	0.5	2	FLG BLIND 1500# RF SS A182 F316/316L	0	0	
610	U04	817	BLIND	5616567	0.75	2	FLG BLIND 1500# RTJ SS A182 F316/316L	0	0	
620										
620							***** ORIFICE FLANGES *****			
630										
630	U04	817	BLIND	5619789	4	4	FLG COMPACT BLIND 1500# RJ SS A182 F316/316L LINE CLASS 1500P1	0	0	
640										
640			ORIFICE FLG	5641520	2	2	WN ORIF SET 1500# RF SS A182 F316/316L SCH 160 BORE	0	0	
650			ORIFICE FLG	5640613	2	2	WN ORIF SET 1500# RTJ SS A182 F316/316L SCH 160	0	0	
660										
660							***** PIPE NIPPLES *****			
670										
670		812		0			NIPPLE BBE CUT FROM PIPE			
680										
680							***** GASKETS *****			
690										
690	U04		GASKET	5672251	0.5	6	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
700	U04	817	GASKET	5673490	4	4	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 1500P1	0	0	
710	U04	817	GASKET	5674479	0.75	2	GASKET 1500# RTJ SS OCTAGONAL RING	0	0	
720	U04		GASKET	5674480	0.75	2	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
730										
730							***** BOLTING *****			
740										
740			BOLTING	5675770	0.5	6	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
750										
750							***** BLINDS & SPACERS *****			
760										
760			SPEC BLIND	5749757	1	2	SPECTACLE BLIND 1500# RF SS A240 GR 316	0	0	
770										

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